Mitigating Cross-Modality Interference in Remote Sensing Detection With Uncertainty-Aware Mixture of Experts

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***Abstract*—Recent years have witnessed rapid progress in multimodal remote sensing datasets and object detection tasks. Representative approaches utilize grid-level sparse mixture-of-experts (MoE) frameworks to accommodate multiple modalities within a unified model, achieving strong performance across various benchmarks. However, existing methods typically employ a shared router to allocate experts across all modalities, which introduces cross-modal interference in gating gradients and leads to unstable expert selection. Moreover, uniform top-*k* routing overlooks situations where the router is uncertain about expert assignment, causing some tokens to miss access to essential expert knowledge. In this work, we introduce a novel MoE framework that integrates modality-specific routers and an uncertainty-aware routing strategy to mitigate cross-modal interference and improve expert utilization. Specifically, modality-specific routers are designed to operate over a shared expert pool, isolating cross-modal gating gradients and decision spaces while maintaining knowledge sharing. Furthermore, we introduce an uncertainty-guided dual routing strategy with normalized entropy, enabling uncertain tokens to engage all experts to acquire comprehensive modality-related knowledge, while the remaining tokens adopt standard top-*k* routing to ensure efficient and reliable expert selection. Extensive experiments on the SOI-Det benchmark demonstrate that our proposed method consistently surpasses previous state-of-the-art approaches across various model scales.**

***Index Terms*—Mixture of experts (MoE), modality-specific routers, remote sensing object detection, uncertainty-aware token training.**

I. Introduction

REMOTE sensing object detection [1], [2], [3], [4], [5], [6], [7], [8], [9], [10] aims to identify and localize targets of interest in aerial and satellite imagery. It is a core capability of remote sensing understanding, with a wide range of real-world applications, including environmental monitoring, forest fire monitoring, and military deployment [11], [12], [13]. In recent years, with the development of remote sensing sensors, modality-specific detection models have emerged for optical image [14], [15], synthetic aperture radar (SAR) image [16], [17], [18], [19], infrared (IFR) image [20], [21], [22], and multispectral image [23], [24], [25].

In practical remote sensing applications, data acquired from different modalities usually exhibit pronounced distributional discrepancies. In this context, modality-specific detection models demonstrate insufficient cross-modality generalization capabilities. Therefore, these models lack practical application value in real multimodal remote sensing scenarios. To enhance the generalization ability of the model, recent methods typically employ multibranch architectures [27], attention architectures [28], and feature fusion architectures [29] to achieve collaborative training with multisource remote sensing data. However, these methods depend on the same scene and aligned paired data, which constrains their scalability in the real world.

To address this limitation, recent studies [26], [30] have explored unified multimodal detection frameworks with public and unaligned datasets. Among them, Li et al. [26] introduce the multimodal and multitask remote sensing object detection benchmark (M2Det), including multimodal datasets and support predefined detection tasks, which include multimodal datasets and predefined detection tasks such as horizontal and oriented bounding boxes. In addition, they further propose a grid-level sparse mixture-of-experts (MoE) framework (SM3Det) coupled with a consistency and synchronization optimization strategy to coordinate training across different modalities. This approach effectively enhances cross-modal feature alignment and overall model generalization. In a related multimodal remote sensing dataset and task settings, RingMoE [30] proposes a structured MoE design with dedicated expert groups for intramodal representation learning, intermodal modeling, and modality-agnostic knowledge sharing. This design enables RingMoE to capture modality-specific knowledge alongside cross-modal dependencies without explicit alignment, achieving superior performance across various remote sensing tasks.

While these unified MoE-based frameworks demonstrate promising performance, they still rely on a shared-routing strategy and uniform sparse activation during training, which

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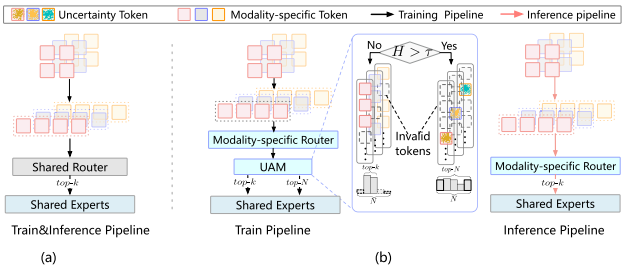


Fig. 1. Comparison of different routing strategies in a grid-level MoE detector. (a) Single shared router over a common expert pool may induce cross-modal gradient interference and unstable expert activation [26]. (b) Proposed UA-MoE employs modality-specific routers over a shared expert pool with an uncertainty-aware training strategy to promote expert exploration. During training, the UAM is applied to all modalities to identify high-entropy (uncertain) tokens, which are routed to all N experts (i.e., top-N branch), while low-entropy (certain) tokens follow the standard top-k routing (i.e., top-k branch). During inference, the UAM and top-N are disabled, and all tokens are routed exclusively through the standard top-k experts.

leaves fundamental challenges in cross-modal interference and expert utilization unresolved. Specifically, existing unified MoE detectors typically adopt a single shared router for all modalities, as shown in Fig. 1(a). However, differences in imaging strategies cause the gating gradients to compete on shared parameters, making the router compromise between heterogeneous modalities and leading to unstable expert selection. Moreover, most existing methods uniformly apply top-k sparse routing during training and inference, overlooking the varied routing difficulty among tokens. However, remote sensing images often contain extensive backgrounds, artifacts, and small objects with large-scale variations. In these regions, routing entropy tends to be high, and assignments are unstable. A few experts are thus driven to overfit uncertain tokens beyond their specialization, causing expert bias and leaving others underutilized. Such an imbalance hinders the development of expert specialization, resulting in unstable convergence and a reduced effective capacity for unified modeling.

Recent MoE research [31], [32] has introduced task-specific and modality-specific routers to mitigate cross-task and cross-modal interference. These methods progressively freeze specialized routers and frequently activate experts to preserve prior knowledge in continual learning, causing subsequent routers to naturally avoid frozen experts and indirectly bypass uncertain tokens. However, in remote sensing multimodal object detection, where modalities are jointly trained, the issue of handling high-uncertainty tokens remains unresolved. Among recent efforts, GW-MoE [33] draws on Global Workspace Theory to identify tokens with high routing uncertainty and broadcast them to all experts during fine-tuning, thereby reducing routing sensitivity. While insightful, this approach faces a key limitation: GW-MoE is tailored to natural language, where broadcasting the uncertainty token to all experts can encourage the acquisition of general knowledge. In multimodal settings, however, the uncertainty distributions differ markedly across modalities. Naively broadcasting the token causes the high-uncertainty modality to dominate expert updates.

To address the above challenges, we propose an uncertainty-aware MoE (UA-MoE) framework for unified multimodal object detection from unaligned data sources. Our UA-MoE consists of two core components: modality-specific routers and an uncertainty-aware training strategy, as illustrated in Fig. 1(b). In contrast to previous approaches, our design explicitly disentangles routing decisions across modalities and performs uncertainty modeling within each modality, aiming to promote more stable expert assignment and clearer specialization under heterogeneous modality distributions. Specifically, we build a dedicated router group on the grid-level MoE architecture. The modality-specific routers decouple routing parameterization from the shared expert pool by assigning an independent router to each modality. This structural separation reduces cross-modal coupling in routing decisions and alleviates potential instability arising from heterogeneous gating signals. Meanwhile, an uncertainty-aware training strategy is introduced, which routes tokens into two processing streams within the shared expert pool. Specifically, an uncertainty-aware module (UAM) is employed to measure the normalized routing entropy of each token. Tokens with high entropy (i.e., uncertain tokens) are broadcast to all N experts to enable broader expert consultation and mitigate unstable assignments, while low-entropy tokens are processed using the standard sparse top-k routing strategy. During inference, the UAM is disabled and all tokens are routed exclusively through the top-k mechanism, introducing no additional computational overhead. Our contributions can be summarized as follows.

1) We introduce a unified UA-MoE framework with modality-specific routers for multimodal remote sensing detection, built upon the existing grid-level sparse MoE to enhance generalization and overall performance.
2) We propose a UAM that leverages normalized routing score entropy to identify high-uncertainty tokens, enabling modality-level knowledge sharing and stabilizing expert specialization.
3) Extensive experiments over three heterogeneous modalities and different detection formats show consistent accuracy gains and more stable routing performance.

II. Related Work

A. Multisource Objection Detection

Multisource object detection integrates complementary information from heterogeneous sensors to improve robustness under complex environments [29], [34], including more recent efforts that enhance robustness under noisy annotations in remote sensing detection [35]. Existing approaches typically perform fusion at the feature, image, or decision level. Feature-level methods learn modality interactions through joint representation learning, such as the dual-stream attention backbone of Meng et al. [36] as well as recent CNN- and Transformer-based RGB-infrared fusion frameworks [37]. Beyond attention-based fusion, convolutional neural network frameworks with explicit cross-modality reconstruction have also been explored, such as CCR-Net [38], which learns compact multimodal representations by enabling effective information exchange across heterogeneous modalities. Image-level fusion synthesizes multispectral inputs before detection, exemplified by Ye et al. [39] through

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complementary structure–texture decomposition. Decision-level strategies merge predictions from different modalities, as in Ahmed et al. [40] via confidence-driven box refinement. While effective, these pipelines rely heavily on paired multi-source data, which is difficult to obtain at scale and limits applicability to real-world deployments with sensor asynchronicity or missing modalities, especially under misaligned SAR–optical settings [28]. To alleviate this dependence, SM3Det [26] introduces the unpaired multimodal detection paradigm, demonstrating that detectors can be trained without aligned cross-modal inputs. However, current unpaired frameworks still provide only coarse cross-modal interaction and remain constrained by their modality-specific designs. In contrast, our work aims to further extend the unpaired setting by enabling more effective cross-modal learning without requiring aligned data, yielding a more scalable and flexible solution.

B. Multidataset Object Detection

Multidataset object detection aims to learn transferable, general-purpose representations by training a single unified model on data from diverse sources and domains. Early studies confirm that joint training across multiple datasets acts as a powerful form of data augmentation, often surpassing the performance of individually trained models [41], [42]. A common paradigm involves training a shared backbone architecture while accommodating dataset-specific differences through dataset-specific output heads. For instance, UniDet [43] successfully implements this by utilizing a shared architecture coupled with distinct, dataset-specific heads. Furthermore, some approaches introduce attention mechanisms to handle domain-specific features, such as the DA [44] network, which employs domain-specific attention layers to adapt to individual datasets. Universal-RCNN [45] integrates features from partitioned detectors using an interdataset graph-based attention module. However, this task becomes significantly more challenging when applied to multimodal remote sensing, where data originates from heterogeneous sensors. To tackle this specific challenge in remote sensing, SM3Det [26] introduces a unified detection model, which employs a grid-level sparse MoE backbone with a shared expert pool, aiming to capture cross-modal commonalities while preserving modality-specific differences. This is complemented by a dynamic learning-rate scheduling strategy to balance the optimization process across different modalities and tasks. Despite these advances, heterogeneity across modalities in a unified training paradigm still leads to significant cross-modal interference and training instability.

C. Mixture of Experts

The MoE paradigm originates from the concept of local experts and gating, where input is routed to specialized experts based on an adaptive gating strategy. This architecture is revitalized in deep learning with the sparsely gated MoE layer [46]. This innovation demonstrates that models could achieve massive parameter capacity by conditionally activating only a small subset of experts for each input. While maintaining a manageable computational cost, thus achieving routing and load balancing. While MoE proves its ability to scale large language models to trillions of parameters [47], the core benefit of conditional computation is quickly adapted to computer vision. For instance, V-MoE [48] demonstrates that replacing the feed-forward network (FFN) blocks in a ViT with sparse experts achieves competitive performance at an inference cost comparable to dense models, while enabling per-sample adaptive computation. In the context of the multimodal remote sensing detection task, the severe distributional heterogeneity and task variation align perfectly with the conditional computation and expert specialization capabilities of MoE. However, conventional MoE routing strategies may introduce cross-modal interference when processing multimodel remote sensing images. As also observed in recent studies that address routing instability and expert imbalance via router freezing or dynamic expert allocation [49]. Building on these works, we introduce UA-MoE to specifically address these challenges, preserving the efficiency of sparse inference while enhancing training stability across modalities.

D. Uncertainty-Aware Strategy

Uncertainty-aware strategies have been widely explored in object detection to stabilize training on difficult or ambiguous inputs [50]. A common approach in dense detectors involves modeling prediction uncertainty. Related work [51] on hyperspectral images further improves robustness by explicitly modeling disentangled priors for background and anomalous components to enhance generalization under complex backgrounds. This includes hard-example mining [52], which reweights losses based on sample confidence, and probabilistic modeling [53], which directly estimates the uncertainty of bounding box localization. In addition, Bayesian detection frameworks such as Bayes R-CNN [54] further model prediction uncertainty to mitigate overfitting and improve reliability on high-resolution imagery. While these methods address uncertainty in the final model output, the MoE introduces a distinct challenge, routing uncertainty. Routing uncertainty arises when the MoE router struggles to make a definitive expert assignment, often due to flat gating scores. This instability can degrade expert specialization and hinder convergence. Several studies have focused on mitigating this problem. For example, moreover, sparse MoE routers often suffer from instability and oscillation. To solve this problem, StableMoE reduces routing variance through two-stage distillation and router freezing [55]. Harder Tasks Need More Experts [56] allocates the number of experts dynamically using difficulty or confidence as a signal, removing the constraint of a fixed top-k. GW-MoE [33] detects routing-uncertain tokens and activates all experts for them during fine-tuning, which reduces sensitivity to a single routing decision without increasing inference cost. Inspired by these strategies that target MoE routing instability, we propose a token uncertainty-aware training strategy. We utilize the normalized entropy of router logits to measure routing ambiguity, then identify and differentially process high-uncertainty tokens during training.

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III. Preliminary

A. Problem Formulation

The core objective of unified multimodal remote sensing detection is to develop a unified model that can accommodate input from heterogeneous modalities and perform detection across tasks and datasets. Given a training set $\mathcal{D}$ composed of different datasets from $\mathcal{M}$ modalities

$$\mathcal{D} = \bigcup_{m=1}^{\mathcal{M}} \mathcal{D}_m, \qquad \mathcal{D}_m = \{(I, y)\} \tag{1}$$

where $\mathcal{M}$ represents the number of datasets or modalities, $\mathcal{D}_m$ is the datasets from the m modality, and I, y denotes the input image and corresponding category label for detection. To enable joint training across different modalities, each dataset-specific category space $\mathcal{C}_m$ is first mapped to a unified global category space $\mathcal{C}$. Then, optimizing a unified detection model f_θ, parameterized by θ, such that it can predict the correct category and bounding box for an input image I. The objective of the learning process is achieved by minimizing the following training loss:

$$\min_{\theta} \ \mathbb{E}_{(I,y,m)\sim\mathcal{D}} \left[\ \mathcal{L}_{\text{det}} \left(f_\theta \left(I \mid m \right), y \right) \ \right] \tag{2}$$

where $\mathcal{L}_{\text{det}} = \mathcal{L}_{\text{branch-1}} + \mathcal{L}_{\text{branch-2}}$ is the detection loss, which is composed of branch-1 and branch-2 loss components. Specifically, the branch one loss is adopted for the SAR modality, which can be defined as

$$\mathcal{L}_{\text{branch-1}} = \mathcal{L}_{\text{QFL}} + \mathcal{L}_{\text{DFL}} + \mathcal{L}_{\text{GIoU}} \tag{3}$$

where $\mathcal{L}_{\text{QFL}}$, $\mathcal{L}_{\text{DFL}}$, and $\mathcal{L}_{\text{GIoU}}$ represent quality focal loss, distribution focal loss, and generalized IoU loss, respectively. In addition, the branch two loss is utilized for the RGB and IFR modalities, defined as

$$\mathcal{L}_{\text{branch-2}} = \mathcal{L}_{\text{RPN-cls}} + \mathcal{L}_{\text{RPN-reg}} + \mathcal{L}_{\text{RoI-cls}} + \mathcal{L}_{\text{RoI-reg}} \tag{4}$$

where $\mathcal{L}_{\text{RPN-cls}}$, $\mathcal{L}_{\text{RPN-reg}}$, $\mathcal{L}_{\text{RoI-cls}}$, and $\mathcal{L}_{\text{RoI-reg}}$ represent the classification and regression losses from the region proposal network (RPN) and the region of interest (RoI) head.

B. Baseline

Remote sensing data exhibit a significant modality gap. Employing a dense model with fully shared parameters leads to mutual interference between the feature representations of different modalities in the shared feature space. Moreover, the different modalities and detection tasks further add complexity to model learning. In a joint training paradigm, this induces mismatched optimization rates and conflicting gradient directions across components. To address these challenges, SM3Det establishes a comprehensive multimodal dataset encompassing SAR, optical, and infrared imagery. Subsequently, the framework proposes targeted solutions from model architecture and model optimization.

1) Grid-Level Sparse MoE: To preserve modality-specific representations while still enabling knowledge sharing, SM3Det integrates a grid-level sparse MoE module into the backbone. Unlike conventional MoE designs that route at the image level, SM3Det routes at the spatial grid of the feature map. This finer granularity allows the model to adapt to local structures. Through a dynamic routing strategy, similar spatial features across different modalities are dynamically allocated to specialized experts while permitting distinct experts to capture modality-specific representations.

2) Dynamic Learning-Rate Adjustment: To address the issue of inconsistent optimization across multiple tasks, SM3Det proposes a dynamic learning-rate adjustment strategy. This strategy monitors task-specific losses and applies exclusive learning-rate adjustments to different network components. This approach effectively balances the convergence speeds and ensures the consistency of the optimization directions. The dynamic learning-rate adjustment strategy consists of two parts: task head adjustment and shared backbone adjustment. The task head adjustment uses the ratio between the current loss of each task and the historical loss obtained from its Exponential Moving Average as a measure of the convergence rate. Then, the softmax function with a temperature coefficient is used to reweight the learning rate of each task head. Tasks that converge faster are assigned smaller learning-rate weights, thereby balancing the convergence process of each task. Shared backbone adjustment operates on the probability distributions of current and historical losses of all tasks. It computes the Kullback–Leibler divergence between these distributions to obtain a batch-wise consistency score. This score generates a dynamic scaling factor for the backbone learning rate. When tasks exhibit aligned optimization signals, the backbone rate is increased to accelerate convergence. When signals conflict, the rate is reduced to temper updates and preserve optimization stability.

IV. Method

A. Overview

In this article, we propose a unified framework designed to mitigate cross-modality interference in multimodal remote sensing detection, achieving balanced and superior detection performance across modalities. Our method is built upon the grid-level sparse MoE architecture [26]. The overall framework of our method is shown in Fig. 2.

We use a shared backbone to process the multimodal input and extract features for early stage representation learning. We introduce the UA-MoE module that replaces the original 1×1 convolutional layers at multiple stages of the backbone. The resulting representations are finally passed to multitask detection heads to produce the end predictions. The proposed UA-MoE module comprises two principal designs, namely modality-specific routing and token uncertainty awareness. Specifically, the modality-specific routing is introduced to address the issues of cross-modal gradient competition and interference that typically arise from shared-routing strategies. These individual routing networks possess independent parameters, ensuring that their gating decisions are influenced solely by the modality-specific feature statistics. This approach effectively isolates cross-modality interference in the decision space. Furthermore, we design an uncertainty-aware strategy that addresses the instability of vanilla top-k routing, which treats all tokens uniformly, leading to volatile assignments

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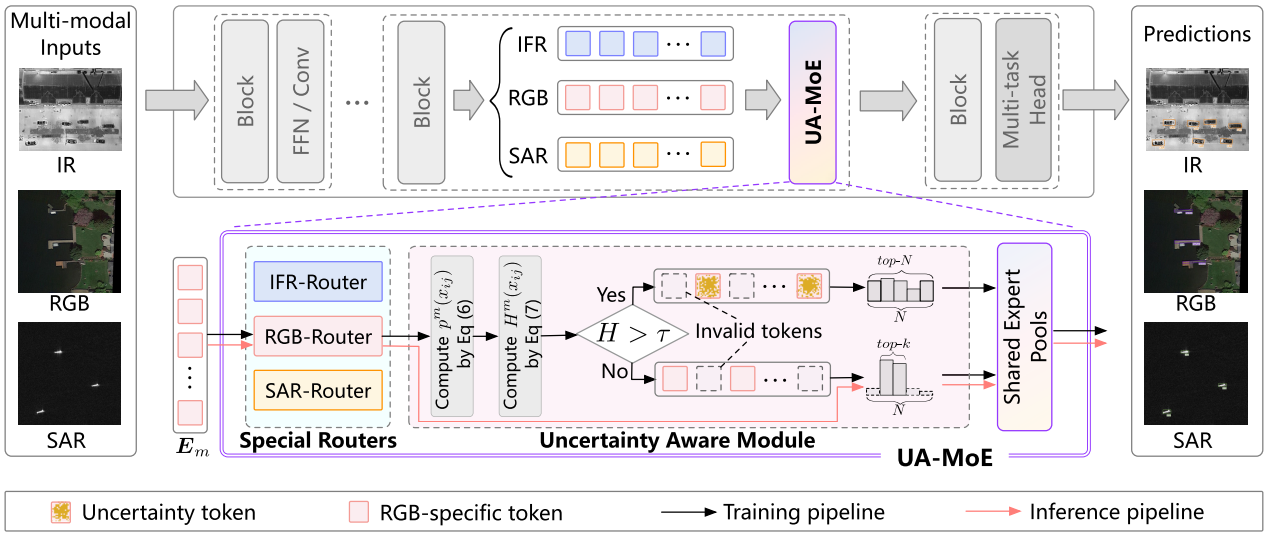


Fig. 2. Architecture of the proposed unified multimodal detector with UA-MoE. The backbone processes images from each modality to extract token embeddings $\boldsymbol{E}_m, m \in \{\text{RGB}, \text{IR}, \text{SAR}\}$. In the diagram, $\boldsymbol{E}_m$ is visualized using RGB tokens for clarity, while the same routing strategy applies to all modalities. Specifically, each $\boldsymbol{E}_m$ is fed into a modality-specific router to produce token-wise gating scores over a shared expert pool. During training (black workflow), the UAM computes the routing distribution $p^m(x_{ij})$ [see (6)] and the normalized routing entropy $H^m(x_{ij})$ [see (7)] of each toke x_{ij} at position (i, j). A threshold $H > \tau$ is applied to split tokens into two routing branches. High-entropy (uncertain) tokens are routed to all N experts (i.e., top-N branch), while low-entropy (certain) tokens follow the standard sparse top-k routing strategy. During inference (orange workflow), the uncertain-aware module and top-N workflow are disabled, and all tokens are routed exclusively through the standard top-k experts, consistent with conventional MoE inference, without introducing additional computational overhead.

Algorithm 1 Training Process

Input: Training set $\mathcal{D}$; initialized model f_θ;
Output: Trained model parameters $f_{\hat{\theta}}$.

for i *in* $[0, iterations]$ **do**
  Sample a mini-batch $\mathcal{B} = \{(I, m)\}$ from $\mathcal{D}$;
  Extract feature embeddings $\boldsymbol{E}_m$ from I;
  for *each token* $x_{ij} \in \boldsymbol{E}_m$ **do**
    Compute modality-specific routing logits $G_m(x_{ij})$;
    Compute probability distribution $p_m(x_{ij})$ by Eq. (6);
    Compute entropy $H^m_{\text{norm}}(x_{ij})$ by Eq. (7);
  for *each modality* m **do**
    Set threshold τ_m by the top-$a\%$ uncertainty entropy $\{H^m_{\text{norm}}(x)\}$;
    $U_m \leftarrow \{x \mid H^m_{\text{norm}}(x) > \tau_m\}$;
    $\mathcal{C}_m \leftarrow \{x \mid H^m_{\text{norm}}(x) \le \tau_m\}$;
    for $x \in C_m$ **do**
      Select $\mathcal{SC}$ by Eq. (10);
      Aggregate expert outputs by top-k;
    for $x \in \mathcal{U}_m$ **do**
      Select $\mathcal{SU}$ by Eq. (11);
      Aggregate all expert outputs;
  Predict $\hat{y}$ and compute $\mathcal{L}_{\text{det}}$ by Eq. (2);

Algorithm 2 Inference Process

Input: Test set $\mathcal{D}_{\text{test}}$; trained backbone $f_{\hat{\theta}}$;
Output: Predictions $\hat{y}$.

Sample a mini-batch $\mathcal{B} = \{(I, m)\}$ from $\mathcal{D}_{\text{test}}$;
Extract feature embeddings E_m from I for each m;
for *each token* $x_{ij} \in \boldsymbol{E}^m$ **do**
  Compute modality-specific routing logits $G_m(x_{ij})$;
  Select $\mathcal{SC}$ by Eq. (10);
  Aggregate expert outputs by top-k;
  Obtain predictions $\hat{y}$ from the detection heads;

for high-uncertainty tokens. We first quantify the uncertainty of each token using normalized routing entropy. During the training stage, tokens exhibiting the highest entropy values are identified. These uncertain tokens are then broadcast to all experts, compelling the experts to learn more robust and comprehensive features. The remaining tokens continue to follow the top-k sparse routing scheme. In the inference phase, all tokens use only the top-k routing, so the uncertainty-aware strategy does not introduce additional computation overhead compared to the standard top-k routing. The procedures of training and inference are detailed in Algorithms 1 and 2, respectively.

B. Uncertain-Token-Aware Training

Multimodal remote sensing images present inherent and significant challenges for object detection. There are significant gaps between the different modal data. Meanwhile, remote sensing images are highly complex, usually encompassing extensive background areas, prevalent artifacts, and scale-variant small objects. The conventional sparse routing treats all tokens uniformly by applying a global top-k strategy, overlooking the key fact that routing difficulty is different across tokens. When tokens arise from ambiguous regions, they significantly hinder the router's ability to make definitive routing decisions. The gating scores become flat or near-uniform across experts. Consequently, the expert assignment for these high-uncertainty tokens becomes highly unstable during the training process.

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Moreover, the constraint of top-k sparse routing prematurely allocates these uncertain tokens to a small subset of experts. This restricted access leads to these tokens being deprived of the opportunity to acquire necessary feature knowledge residing within other experts, thereby constraining effective cross-expert information sharing.

These observations suggest that the distribution of the router's gating scores provides an effective criterion for discriminating high-uncertainty tokens. Building on this insight, we propose a token uncertainty-aware training strategy that identifies the certainty of different tokens and stabilizes the gating process of modality-specific routers, thereby enabling the model to acquire more comprehensive and transferable knowledge. As shown in Fig. 2, this strategy consists of two core components: uncertainty token quantification and uncertainty token global broadcasting.

1) *Uncertainty Quantification:* To mitigate bias introduced by intermodality distributional discrepancies, we compute routing uncertainty independently within each modality. Specifically, for a spatial token $x_{ij} \in \mathbb{R}^C$ from the embeddings $\boldsymbol{E}_m$ of m modality, we first compute its routing logits $G^m(x_{ij}) \in \mathbb{R}^N$ over N experts, defined as

$$G^m\left(x_{ij}\right) = \left[g_0, g_1, \ldots, g_{N-1}\right] \tag{5}$$

where i and j represent the location of the token in the feature map and g_n denotes the routing logits corresponding to the nth expert. Then, we obtain the temperature-scaled routing probability distribution p as follows:

$$p^m\left(x_{ij}\right) = \operatorname{softmax}\left(G^m\left(x_{ij}\right)/t\right) \tag{6}$$

where t is a temperature. To quantify the uncertainty of the routing distribution, we map it to $[0, 1]$ using normalized entropy [57], which can be formulated as

$$H^m_{\text{norm}}\left(x_{ij}\right) = \frac{-\sum_{n=1}^{N} p_n^m\left(x_{ij}\right) \log p_n^m\left(x_{ij}\right)}{\log N} \tag{7}$$

where log denotes the natural logarithm and $p_n^m(x_{ij})$ is the routing probability distribution of the nth router. A higher value of $H^m_{\text{norm}}(x_{ij})$ indicates a more uniform distribution, suggesting that the router is relatively uncertain about which expert should handle the token.

2) *Uncertainty Token Identification and Processing:* To identify uncertain tokens, we adopt a within-modality percentile thresholding strategy. Specifically, for each modality, we regard the top $a\%$ tokens with the highest normalized entropy as uncertainty tokens. Formally, the set of certain tokens in modality m is defined as

$$\mathcal{C}^m = \left\{x_{ij}^m \mid H^m_{\text{norm}}\left(x_{ij}\right) \le \tau\right\} \tag{8}$$

and the set of uncertain tokens is defined as

$$\mathcal{U}^m = \left\{x_{ij}^m \mid H^m_{\text{norm}}\left(x_{ij}\right) > \tau\right\} \tag{9}$$

where τ denotes the minimum normalized entropy value among the top $a\%$ tokens within modality m. This threshold τ is thus a dynamic value for each modality, ensuring that the selection is adaptive yet consistent across different modalities.

During training, we employ a dynamic dual-path routing strategy to handle both certain and uncertain tokens. Tokens identified as uncertain are broadcast to all N experts, encouraging the experts to collaboratively learn representations for these challenging cases. The remaining tokens follow the standard top-k sparse routing protocol, maintaining computational efficiency and preserving expert sparsity for these certain tokens. This joint learning enhances the experts' capacity to capture complex patterns and promotes more stable specialization. In inference, the uncertainty-aware strategy is disabled, and all tokens strictly adhere to the efficient top-k routing. This is because the experts have already acquired sufficient generalization capability during training through the joint learning of uncertain tokens, making the uncertainty-aware process unnecessary at inference. Consequently, our method markedly improves training stability and model robustness without introducing any additional computational overhead during inference.

C. Modality-Specific Routers With Shared Experts

Remote sensing modalities differ substantially in imaging mechanisms, statistical characteristics, and textural appearance. Under a shared-routing scheme, tokens from different modalities backpropagate their gating gradients through the same routing parameters. This introduces gradient competition among modalities, forcing the router to converge on a suboptimal compromise across these heterogeneous distributions. Therefore, the router's expert selection becomes highly unstable, as it cannot simultaneously satisfy the modality-specific requirements. This instability ultimately impedes the generalization capability of the model. To address this challenge, we propose a modality-specific routing with a shared expert pool strategy, as presented in Fig. 2. This architecture is designed to isolate the gating decisions for each modality, thereby eliminating gradient competition while preserving the knowledge-sharing benefits to the shared expert pool.

Given an image from modality m, the backbone produces a multiscale feature map, and the token $x^m_{i,j}$ at grid location (i, j) serves as the basic routing unit. For each modality, an independent router $G^m(\cdot)$ maps tokens to expert logits. We enforce sparsity by selecting the top-k experts and aggregating their outputs for each token of the certainty tokens $\mathcal{C}^m$

$$\mathcal{SC}^m_{ij} = \text{top-}k\ \left(G^m\left(x^m_{ij}\right)\right), x^m_{ij} \in \mathcal{C}^m \tag{10}$$

where $\mathcal{SC}^m_{ij}$ is the index set of the k highest scoring experts. For the uncertain tokens in the set $\mathcal{U}^m$, we activate all experts to collaboratively process these uncertain tokens through the routing strategy

$$\mathcal{SU}^m_{ij} = \text{top-}N\ \left(G^m\left(x^m_{ij}\right)\right), x^m_{ij} \in \mathcal{U}^m \tag{11}$$

where $\mathcal{SU}^m_{ij}$ represents the expert index set activated for uncertain tokens. Subsequently, the outputs from the selected experts are aggregated via normalized gates to produce the final UA-MoE output. The proposed dual-path routing strategy adaptively balances computational efficiency and model robustness. By integrating the uncertainty-aware strategy with modality-specific routing, our approach mitigates routing instability and improves the generalization of the object detection model.

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TABLE I

MODEL PERFORMANCE ON THE SOI-DET DATASET WITH CONVNEXT-T [63] AS THE BACKBONE. EACH ENTRY REPORTS (MAP/@50/@75) FOR THREE DATASETS AND THE OVERALL AVERAGE. WE LABEL THE BEST AND SECOND METHODS WITH BOLD AND UNDERLINE STYLES

Method	SARDet-100K			DOTA			DroneVehicle			Overall		
	mAP	@50	@75	mAP	@50	@75	mAP	@50	@75	mAP	@50	@75
Single-dataset baselines												
UIU-Net [64]	—	—	—	—	—	—	41.01	68.61	45.96	—	—	—
GFL [65]	57.31	87.44	61.99	—	—	—	—	—	—	—	—	—
O-RCNN [66]	—	—	—	45.31	77.70	46.45	46.09	74.78	52.79	—	—	—
Unified detectors												
Simple Joint Training [45]	53.46	84.11	57.29	45.18	76.37	46.78	44.99	73.28	51.50	47.05	77.56	50.11
DA [44]+ConvNext-T	53.86	84.93	58.09	46.23	78.47	47.58	48.21	77.43	56.16	48.37	79.76	51.66
UniDet [43] (Partitioned)	53.81	84.70	57.43	46.49	78.28	48.59	47.99	77.17	55.74	48.47	79.55	52.01
Uncertainty loss [67]	53.43	84.81	57.41	**46.94**	**78.73**	**49.08**	48.78	77.96	56.88	48.79	79.99	52.50
SM3Det [26]	60.64	89.94	65.06	46.47	77.88	48.24	48.87	77.99	56.90	50.20	80.68	53.79
Ours	**62.65**	**90.87**	**67.87**	46.24	77.79	47.85	**49.54**	**78.55**	**58.27**	**50.66**	**80.95**	**54.47**

V. EXPERIMENTS

A. Datasets

To evaluate the proposed method, we conduct experiments on the SOI-Det dataset [26] and an additional custom dataset. The SOI-Det dataset is constructed by merging SARDet-100K [58], DOTA-v1.0 [59], and DroneVehicle [20] datasets. The custom dataset is constructed by merging OGSOD-1.0 [60], DIOR [61], and HIT-UAV [62]. The SARDet-100K dataset is an SAR object detection dataset, consisting of 94 493 images for training and 11 613 images for testing. The DOTA-v1.0 dataset is collected from optical aerial objects. It comprises 25 028 training images and 17 041 testing images. The DroneVehicle dataset is an infrared vehicle detection dataset, which contains 17 990 training images and 8980 testing images. In addition, the OGSOD-1.0 (SAR), DIOR (RGB), and HIT-UAV (IFR) datasets contain 14 664/3667, 5862/5863, and 2008/287 training/test images, respectively.

B. Implementation Details

We follow the baseline SM3Det for data processing and dynamic learning-rate adjustment. The network is trained using the AdamW optimizer with a batch size of 4. To maintain a fair comparison with individually trained models, the total number of iterations for joint training is set to equal the cumulative iterations required for training on each dataset separately. During data sampling, we construct each batch by uniformly sampling from the three modalities of different datasets (SAR, RGB, and IFR) at a 2:1:1 ratio. We evaluate the performance of the proposed method across three backbone scales of ConvNext [63], including ConvNext-T, ConvNext-S, and ConvNext-B, as well as LskNet-T [68]. For computing the threshold τ of uncertain token selection, we set $a\%$ to 10% for ConvNext-T, ConvNext-S, and LskNet-T, and 5% for ConvNext-B. UA-MoE replaces the 1×1 convolutional layers on the even layers in stages 3 and 4 of the ConvNext backbone. The expert pool size is $N = 8$ with top-$k = 2$. As SM3Det, we use standard metrics AP@50, AP@75, and mAP@[0.5:0.95], and report the overall mAP averaged across the three datasets.

C. Main Results

We evaluate the performance of the proposed model under two training paradigms: single-dataset training [64], [65], [66] and simple joint training [45]. We also compare it with four state-of-the-art baselines: DA network [44] implemented within a ConvNext-T backbone [63], UniDet with partitioned detection heads [43], UniDet augmented with the uncertainty loss [67], and SM3Det [26], the baseline for our approach.

The main comparison results are presented in Table I. Our method achieves the best overall performance on SOI-Det with 50.66% mAP, improving over SM3Det by 0.46%. Although the overall gain is moderate in absolute value, it is achieved under the same backbone architectures and training protocols in a unified multimodal detection setting, which is inherently more challenging than single-dataset training. More importantly, this improvement is obtained without increasing inference-time computational complexity. As shown in Table II, UA-MoE reduces FLOPs and parameter count while improving FPS, although it introduces a moderate increase in peak GPU memory due to the additional modality-specific routers. From the dataset-wise perspective, the benefit of UA-MoE is more evident on the modalities that are more affected by cross-modality interference. Specifically, UA-MoE improves mAP by 2.01% on SARDet-100K and by 0.67% on DroneVehicle, indicating better multimodal balance and more stable expert allocation. On the DOTA dataset, UA-MoE yields a small decrease of 0.23% mAP. It is because DOTA is an optical-only benchmark with rich appearance cues and dense texture details, for which the baseline shared-router framework can already adapt well to the characteristics of a single optical modality. In contrast, UA-MoE is designed to improve routing stability and expert allocation under unified optimization across heterogeneous modalities, rather than to overadapt the routing behavior to one specific optical dataset. Therefore, the small decrease on DOTA can be regarded as a minor tradeoff in unified multimodal optimization, rather than a weakness of the proposed design.

Furthermore, the proposed method is compared against training three independent specialist models (three models), which utilize the GFL [65] model for the SARDet-100K

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TABLE II

EFFICIENCY COMPARISON OF OURS AND COMPARED METHODS ON THREE REMOTE SENSING BENCHMARKS DURING INFERENCE, IN TERMS OF COMPUTATIONAL COMPLEXITY (FLOPS), MODEL SIZE (#P), INFERENCE SPEED (FPS), AND PEAK GPU MEMORY CONSUMPTION (MB)

Method	FLOPs (G)	#P (M)	SARDet-100K		DOTA		DroneVehicle	
			FPS ↑	GPU (MB) ↓	FPS ↑	GPU (MB) ↓	FPS ↑	GPU (MB) ↓
Single-dataset baselines								
GFL [65]	131	36	14.11	313.96	—	—	—	—
O-RCNN [66]	136	45	—	—	11.67	430.74	—	—
O-RCNN [66]	136	45	—	—	—	12.72	419.07	—
Unified detectors								
Simple Joint Training [45]	403	66	34.09	595.75	25.68	758.78	29.03	736.49
DA [44]+ConvNext-T	403	66	23.34	597.30	12.37	760.22	13.42	737.97
UniDet [43] (Partitioned)	403	66	15.55	595.75	12.73	755.42	13.96	736.30
Uncertainty loss [67]	403	66	12.89	595.75	10.50	752.49	21.37	736.14
SM3Det [26]	487	178	22.11	1024.75	18.22	1185.41	19.72	1165.52
Ours	472	176	25.54	1108.85	22.54	1501.94	23.98	1602.09

TABLE III

ABLATION ACROSS BACKBONE SCALES. EACH ENTRY REPORTS (MAP/@50/@75) FOR THREE DATASETS AND THE OVERALL AVERAGE. WE LABEL THE BEST AND SECOND METHODS WITH BOLD AND UNDERLINE STYLES

Backbone	Method	SARDet-100K			DOTA			DroneVehicle			Overall		
		mAP	@50	@75	mAP	@50	@75	mAP	@50	@75	mAP	@50	@75
ConvNext-B	Baseline	65.20	92.41	**70.02**	45.86	76.35	48.10	51.09	**80.07**	60.34	51.33	80.77	55.51
	Baseline + Uncertain	65.18	92.31	69.71	46.18	**77.46**	48.50	51.05	79.94	60.46	51.50	**81.36**	55.69
	Ours	**65.32**	**92.46**	**70.02**	**46.45**	77.32	**48.82**	**51.12**	79.94	**60.75**	**51.70**	81.32	**56.01**
ConvNext-S	Baseline	62.98	91.49	68.77	45.33	76.03	46.98	49.89	78.79	58.88	50.28	80.13	54.03
	Baseline + Uncertain	63.34	91.38	68.71	46.36	**77.38**	48.61	50.29	**79.30**	59.10	51.03	**80.98**	55.27
	Ours	**63.93**	**91.51**	**68.93**	**46.74**	77.30	**49.37**	**50.54**	79.20	**59.71**	**51.44**	80.94	**55.87**
ConvNext-T	Baseline	60.64	89.94	65.06	**46.47**	**77.88**	**48.24**	48.87	77.99	56.90	50.20	80.68	53.79
	Baseline + Uncertain	62.43	**90.92**	66.96	45.80	76.76	48.10	48.89	77.72	57.36	50.23	80.21	54.23
	Ours	**62.65**	90.87	**67.87**	46.24	77.79	47.85	**49.54**	**78.55**	**58.27**	**50.66**	**80.95**	**54.47**

dataset and the O-RCNN [66] model for both the DOTA and DroneVehicle datasets. The proposed unified detector achieves higher overall performance, raising mAP from 48.23% to 50.66%. This result shows that the proposed unified model not only avoids the overhead of maintaining multiple models but also gains stronger generalization through cross-modal knowledge sharing. In addition to these general-purpose specialist detectors, we further include a comparison with UIU-Net [64], a recent state-of-the-art infrared-oriented detection framework, which can be regarded as a modality-specific specialist model for IFR imagery. Although UIU-Net is originally designed for binary infrared target detection, we adapt it to a multiclass setting and evaluate it on the DroneVehicle dataset. As reported in Table I, UIU-Net also demonstrates strong detection capability on multiclass IFR targets, while our unified model achieves higher overall performance under the same setting.

Our method consistently outperforms other mainstream unified training methods in overall mAP. Compared to the second-best method, Uncertainty loss, our approach still delivers a 1.87% performance gain. These results indicate that combining modality-specific routers with the uncertainty-token-aware training strategy effectively mitigates cross-modality interference and stabilizes routing decisions in a unified detection framework. To further validate the scalability of our method, we conduct comparative experiments across different ConvNext backbone scales, as detailed in Table III. The results demonstrate that the proposed method consistently outperforms the SM3Det across all backbone scales. Specifically, our method improves overall mAP by 0.46% on ConvNext-T, by 1.16% on ConvNext-S, and by 0.37% on ConvNext-B.

D. Generalization to Other Datasets and Backbones

To further evaluate robustness and generalization, we extend our experiments to multiple public datasets and backbone architectures. Specifically, we introduce three widely used benchmarks covering different sensing modalities and imaging conditions, including OGSOD-1.0 (SAR), DIOR (RGB), and HIT-UAV (IFR). As shown in Table IV, the proposed method consistently outperforms existing unified detection methods, including the recent state-of-the-art approach SM3Det. Under the ConvNext-T backbone, our method improves the overall mAP from 39.50% to 45.20% (+5.7%), with notable gains on HIT-UAV (+0.5% mAP) and OGSOD-1.0 (+7.6% mAP). To further verify backbone generalization, we replace ConvNext-T with a recent architecture, LskNet-T, while keeping all other settings unchanged. Although LskNet-T does not yield higher performance than ConvNext-T, our method still achieves a clear improvement over the corresponding SOTA baseline, increasing the overall mAP from 39.70% to 40.33% (+0.63%).

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TABLE IV

GENERALIZATION TO OTHER DATASETS AND BACKBONES. THE EVALUATED DATASETS INCLUDE OGSOD-1.0 [60] (SAR), DIOR [61] (RGB), AND HIT-UAV [62] (IFR), AND THE BACKBONE NETWORKS INCLUDE CONVNEXT-T [63] AND LSKNET-T [68]. EACH ENTRY REPORTS (MAP/@50/@75) FOR THREE DATASETS AND THE OVERALL AVERAGE. WE LABEL THE BEST AND SECOND METHODS WITH BOLD AND UNDERLINE STYLES

Method	OGSOD-1.0			DIOR			HIT-UAV			Overall		
	mAP	@50	@75	mAP	@50	@75	mAP	@50	@75	mAP	@50	@75
ConvNext-T												
Simple Joint Training [45]	32.10	66.23	26.74	36.35	63.06	38.80	35.53	67.68	35.71	35.75	64.22	36.96
DA [44]+ConvNext-T	31.58	67.65	26.25	39.37	66.76	43.18	37.89	68.91	34.07	38.27	67.24	39.74
UniDet [43] (Partitioned)	39.43	75.04	36.42	37.41	65.04	39.08	38.48	69.67	37.78	37.82	66.94	38.56
Uncertainty loss [67]	39.28	75.62	36.35	39.02	65.66	41.90	36.75	64.96	38.15	38.64	66.60	40.64
SM3Det [26]	42.33	80.82	39.14	39.06	66.70	41.79	39.58	**73.40**	**41.43**	39.50	69.41	41.44
Ours	**49.93**	**85.43**	**48.24**	**45.77**	**71.03**	**49.68**	**40.08**	67.07	38.68	**45.20**	**71.87**	**47.56**
LskNet-T												
Simple Joint Training [45]	25.83	55.06	21.36	41.12	66.63	44.15	40.04	72.08	40.85	39.29	66.36	41.12
DA [44]+LskNet-T	25.50	53.98	21.39	39.01	63.97	41.91	38.10	65.13	39.05	37.40	63.11	39.20
UniDet [43] (Partitioned)	27.75	56.51	**24.61**	38.33	62.86	41.06	39.47	66.93	40.80	37.40	62.91	39.25
Uncertainty loss [67]	24.64	52.15	20.11	35.29	60.74	37.43	36.04	65.60	36.80	34.28	60.69	35.46
SM3Det [26]	26.41	55.86	22.00	41.52	**66.74**	44.72	40.38	68.70	40.32	39.70	65.92	41.50
Ours	**27.93**	**58.11**	23.68	**41.98**	66.41	**45.50**	**41.15**	**73.62**	**45.92**	**40.33**	**66.81**	**43.24**

TABLE V

ABLATION STUDY OF MODALITY-SPECIFIC ROUTER ON CONVNEXT-T BACKBONE. EACH ENTRY REPORTS MAP FOR THREE DATASETS AND THE OVERALL AVERAGE

Method	SARDet-100K	DOTA	DroneVehicle	Overall
Ours w/o modality-specific router	62.43	45.80	48.89	50.23
Ours with modality-specific router	**62.65**	**46.24**	**49.54**	**50.66**

TABLE VI

TRAINING COST ANALYSIS OF TOP-N BRANCH FOR A SINGLE EPOCH

Method	Time ↓	s/it ↓	GPU ↓
Ours w/o UAM	3262.06	1.17	10057.78
Ours with UAM	3359.51 (+2.99%)	1.20 (+2.99%)	10376.50 (+3.17%)

E. Qualitative Results

We present a qualitative comparison of our methods and other methods across three remote sensing modalities datasets, as shown in Fig. 3. To provide a comprehensive evaluation, we present two representative examples for each modality.

In the first SAR modality, where strong speckle noise is present (first row), all comparison methods fail to detect the small ship in the top-right corner, while our method successfully detects all objects. In the more complex coastal scene (second row), several methods (e.g., single-dataset training, simple joint training, DA, UniDet, and Uncertainty loss) miss some small ships; SM3Det also shows missed detections and redundant boxes. The third and fourth rows provide the detection results on the DOTA dataset. As shown in the third row, all methods can detect the big harbor on the left, whereas our method better identifies challenging objects such as the tiny harbor on the right. In addition, DA, UniDet, Uncertainty loss, and SM3Det misclassify the container as a large vehicle in the fourth row. The last two rows highlight object confusion on the DroneVehicle dataset. Compared with the comparison methods that suffer from false positives between truck and freight car, the proposed method benefits from the UA-MoE module and modality-specific routing, which reduce cross-modality interference and improve detection performance.

F. Grid-Level Experts Visualization

In this section, we visualize the top-1 expert assignments of a trained ConvNext-T model across MoE stages (Stages 3 and 4) and compare UA-MoE with UA-MoE *w/o* modality-specific routing and SM3Det in Fig. 4. We use the same inputs, inference settings, and a fixed color palette for the eight experts to ensure a fair comparison. The results highlight two advantages of UA-MoE in routing stability and expert specialization.

First, our method exhibits stronger object-level semantic consistency. In SM3Det, tokens within the same object are often routed to multiple experts (e.g., aircraft/containers in RGB), indicating fragmented assignment. In contrast, UA-MoE tends to assign tokens from the same target to a more consistent dominant expert, and this tendency becomes particularly evident in Stage 4 Block 2. In these deeper layers, routing is driven less by local texture variations and more by modality-relevant high-level semantics, so the expert assignments within the same modality become more consistent. With modality-specific routers, routing decisions are decoupled across modalities while the expert pool remains shared, which reduces cross-modality interference in the routing space. Together with the uncertainty-aware training strategy, this results in more stable expert allocation for deep features with similar semantic roles. Consequently, the concentrated routing pattern in deep layers reflects more semantically coherent expert assignment under unified multimodal training, rather than a direct loss of feature discriminativeness.

Second, this interpretation is further supported by the ablation in Fig. 4(b). When modality-specific routing is removed, the deep-layer expert assignments become more diverse and fragmented, resembling the shared-router behavior of SM3Det. However, as shown in Table V, this variant leads to consistent degradation in the overall mAP as well as in most dataset-level metrics. These results indicate that visually stronger expert differentiation in deep layers is not necessarily accompanied by better detection performance. Instead, in the

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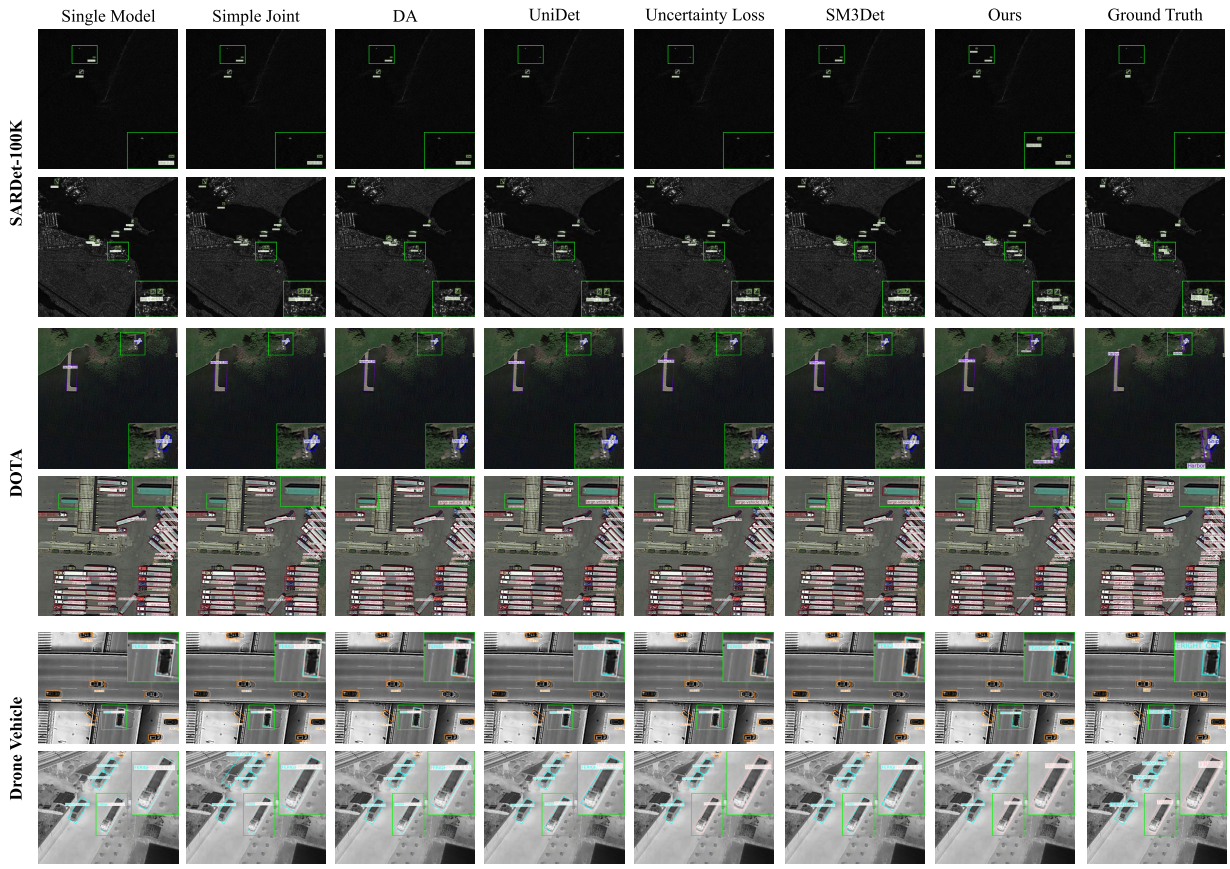


Fig. 3. Qualitative comparison across three remote sensing modalities: SAR, optical RGB, and infrared. From left to right, the eight columns show qualitative results for single model, simple joint training [45], DA [44], UniDet [43], uncertainty loss [67], SM3Det [26], Ours, and ground truth. Our method yields fewer misses and false positives across all modalities, indicating reduced cross-modality interference and improved consistency.

unified multimodal setting, such fragmented routing patterns are more likely to reflect less stable expert allocation under cross-modality interference. This instability weakens the semantic consistency of expert utilization and ultimately limits the effectiveness of unified multimodal representation learning.

G. Computational Efficiency

We further evaluate the computational efficiency and running time of the proposed method by measuring FLOPs, parameter size (#P), inference speed (FPS), and peak GPU memory (GPU), with results reported in Table II. It can be observed that the proposed method reduces #P by 1.12% and FLOPs by 3.08% compared to SM3Det. Consequently, our model achieves higher FPS across all three datasets. For example, on the SARDet-100K dataset, FPS is improved by 15.51%. These gains are primarily attributable to inserting fewer MoE blocks than SM3Det, and the uncertainty-token-aware training strategy is disabled during inference, imposing no additional deployment overhead. However, we observe higher peak GPU memory, which is expected because modality-specific routing maintains distinct gating operations for each modality. In summary, our method offers a favorable tradeoff: a moderate increase in peak GPU memory yields higher FPS with lower FLOPs and fewer parameters.

In addition to inference-time efficiency, we evaluate the training-stage overhead introduced by the uncertainty-aware top-N branch. We compare models trained with and without UAM under identical settings and report the cost per epoch in Table VI. The top-N branch introduces only minor overheads of + 2.99% in training time/per-iteration latency and + 3.17% in GPU memory. Since this branch is fully disabled during inference and improves accuracy, the additional training cost is justified and does not compromise practicality.

H. Analysis of Router Entropy

To further validate the effectiveness of the proposed method in routing decisions, we introduce entropy to quantify the router's uncertainty in assigning experts. Higher entropy indicates greater ambiguity in routing decisions, while lower entropy signifies more definitive assignments.

As shown in Fig. 5, we compare the evolution of token entropy during training between the proposed method and the SM3Det baseline. This analysis includes two views: 1) the total entropy sum across all tokens, reflecting overall routing uncertainty; and 2) the entropy sum for the top-10% highest entropy tokens, reflecting the model's behavior on the most ambiguous inputs. Fig. 5(a) illustrates the modality-specific routing stability. As depicted in the orange curve, SM3Det starts with high total entropy and decreases slowly; by the end of training, its total entropy remains at 230 000, indicating persistent routing uncertainty in the shared router. In contrast, our method begins with much lower total entropy and converges faster, indicating that modality-specific routing reduces

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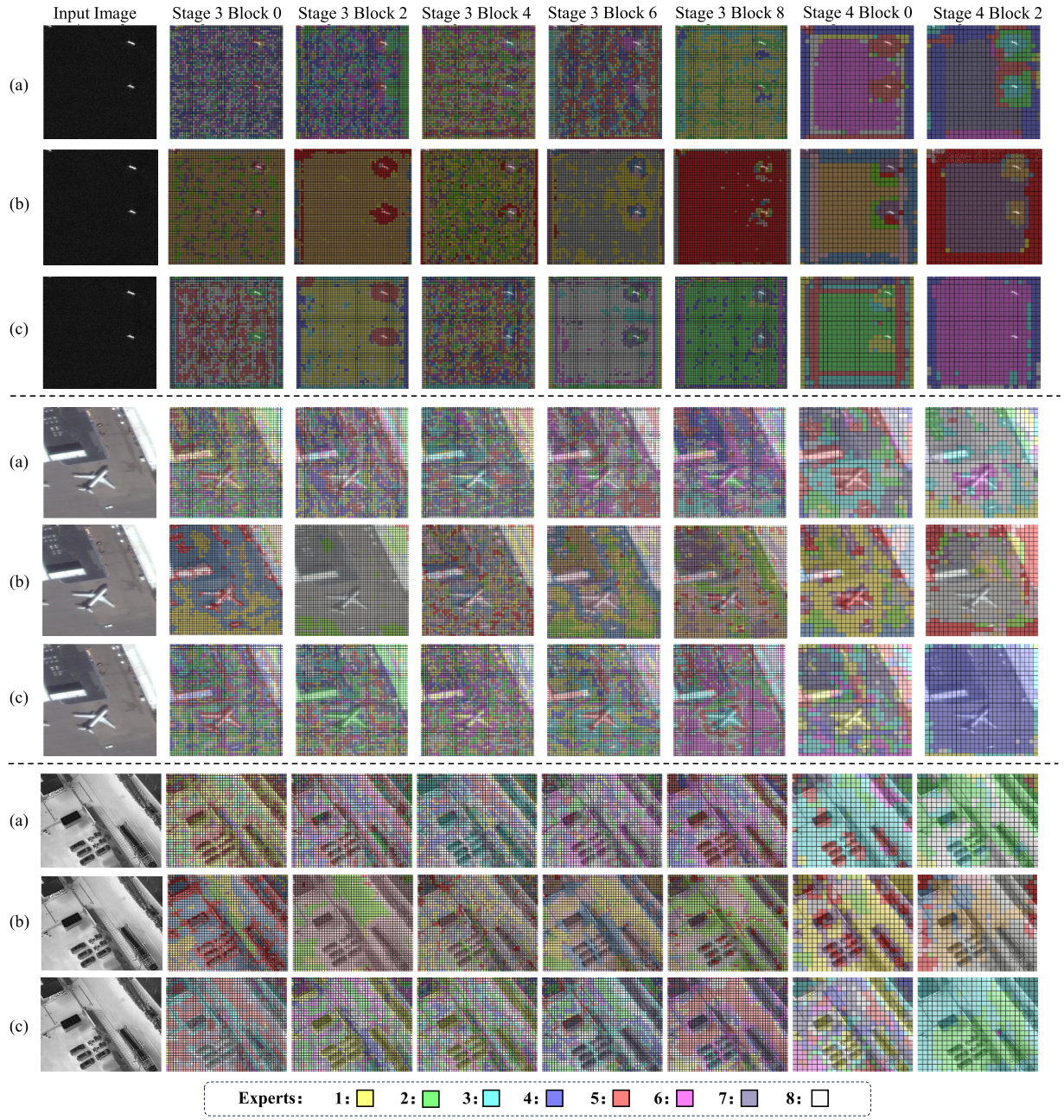


Fig. 4. Visualization of grid-expert activations across all MoE stages for SAR, RGB, and IFR modalities. Rows correspond to SARDet-100K, DOTA, and DroneVehicle datasets. Each square grid represents the receptive field for the specified stage, and different colors indicate local grid regions processed by distinct experts. (a) SM3Det with a shared router. (b) UA-MoE *w/o* modality-specific routing. (c) Proposed method uses modality-specific routers and uncertainty-aware training. The proposed method exhibits enhanced semantic awareness, leading to more stable routing and better expert specialization.

router ambiguity. In addition, Fig. 5(b) validates the proposed uncertainty-aware training strategy by focusing on the most uncertain tokens. While SM3Det struggles to reduce ambiguity for these tokens, our method's entropy drops sharply, showing that broadcasting high-uncertainty tokens to all experts is effective and improves routing stability.

I. Ablation Study and Analysis

In this section, we conduct comprehensive ablation studies on the SOI-Det dataset to validate the effectiveness of our two core components: the uncertainty-token-aware training strategy and the modality-specific routers. As shown in Table III, we evaluate our components across three ConvNext backbone scales (ConvNext-B/S/T) based on the following configurations. "**Baseline**" refers to the SM3Det. "**Uncertain**" incorporates the uncertainty-token-aware training strategy into the baseline model. Finally, "**Ours**" represents the full UA-MoE framework, which integrates the uncertainty-aware training strategy and the modality-specific routers.

1) Effectiveness of Uncertainty-Aware Training: In this experiment, we first evaluate the performance of the "Baseline + Uncertain" model on three ConvNext backbone scales. As shown in Table III, by adding the uncertainty-token aware training strategy to the baseline, the

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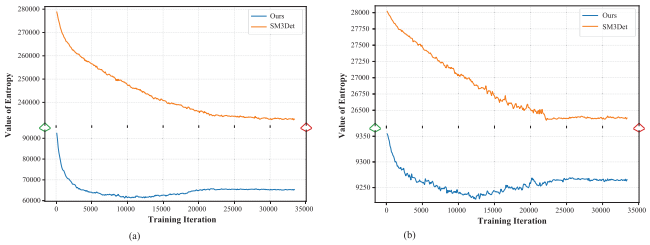


Fig. 5. Comparison of router entropy evolution between our proposed method and SM3Det during training. (a) Sum of entropy for all tokens. (b) Sum of entropy for the top-10% highest entropy tokens. Higher entropy indicates greater routing ambiguity for each token, while lower entropy reflects more confident expert assignments. Our method consistently reduces both overall and top-10% token entropy, demonstrating improved routing clarity and stability.

"Baseline + Uncertain" model achieves consistent gains in overall mAP across all three backbones. Specifically, adding the uncertainty-token-aware training strategy increases overall mAP by 0.17% on the ConvNext-B backbone, 0.75% on the ConvNext-S backbone, and 0.03% on the ConvNext-T backbone.

2) Effectiveness of Modality-Specific Routers: We add the modality-specific routers module to the "Baseline + Uncertain" model on three ConvNext backbone scales. This model demonstrates the synergistic effect of the two components. The addition of modality-specific router modules results in an improvement of the overall mAP by 0.41% on the ConvNext-S backbone. Similarly, the full model further boosts the overall mAP by 0.43% on the ConvNext-T and 0.20% on the ConvNext-B backbone.

VI. Conclusion

In this article, we introduced UA-MoE, an uncertainty-aware MoE framework designed for unified multidataset object detection in remote sensing. To mitigate the unstable expert selection of high-uncertainty tokens, we designed an uncertainty-aware training strategy. This strategy identifies ambiguous tokens using normalized entropy and broadcasts them to all experts only during training, thereby strengthening expert specialization and stabilizing routing without increasing deployment cost. Additionally, we introduce a modality-specific router module, which decouples gating decisions to eliminate the cross-modal interference from the shared router. Comprehensive experiments demonstrate that the proposed UA-MoE consistently achieves superior performance and routing stability across multiple model scales, highlighting the effectiveness of combining uncertainty-guided learning with modality-specific routing.

1) *Limitation:* One limitation of our framework is that the current study lies in the fixed modality-specific router design, which assumes a known and static set of sensing modalities. Extending UA-MoE to dynamically accommodate new or unseen modalities in continual learning or open-domain settings would be an interesting direction for future work. Nevertheless, the proposed framework provides a robust foundation for scalable multimodal detection in remote sensing.

2) *Future Work:* UA-MoE currently assumes a fixed set of sensing modalities with a dedicated router per modality. Future work will extend it to additional and potentially unseen modalities via adaptive router designs. Robustness under missing, corrupted, or degraded modalities will be improved through partial-observation training (e.g., modality dropout) and confidence-aware routing that downweights unreliable modalities and reallocates expert capacity. In addition, more principled uncertainty calibration for routing probabilities and the alignment between routing uncertainty and detection uncertainty will be explored, and UA-MoE will be scaled to larger multisource datasets and expert pools.

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